

算力与账单

谁为 AI 数据中心繁荣买单？

Compute and the bill: who pays for the AI data-centre boom?

Thesis / 主旨

AI 数据中心并不是单纯的“用电大户”。它们是一组对电网、土地、水、地方财政和公共信用的未来索取权。真正的问题不是是否建设，而是每一项成本在哪一张资产负债表上结算。

AI data centres are not merely large consumers of electricity. They are claims on grids, land, water, local fiscal capacity and public credit. The question is not whether to build them, but on whose balance sheet each cost will settle.

Take-aways for stakeholders / 首页导读

读者 / Audience	最重要的问题 / The question that matters most
监管者 / Regulators	成本按 cost causation 分配，还是进入 rate base 由普通用户偿还？
地方政府 / Local governments	扣除 abatements、水、电、道路和退出风险后，净财政账是否仍为正？
居民与社区 / Ratepayers and communities	负债是否落在电费、税收或生活质量上，而收益留在别处？
Utility	新资产是否由可靠付款承诺支持，还是把预测风险转给未来用户？
科技公司与开发商 / Tech firms and developers	“自己付钱”的证据应是 tariff、collateral、minimum bill、exit fee 和互联协议。
投资者与债权人 / Investors and creditors	风险在于公共基础设施付款能力低于私人负荷预测。
记者与研究者 / Journalists and researchers	画 use-source account，找出 residual risk bearer。

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1 A Boom With A Bill / 繁荣背后的账单

A concrete question / 具体钩子

当一家 AI 数据中心承诺“我们会自己付电费”时，真正的问题不是它是否购买电力，而是变电站、输电升级、峰值容量、备用资源、退出风险和地方税收减免分别由谁付款。

When an AI data centre promises to pay for its own electricity, the real question is not whether it buys power. It is who pays for substations, transmission upgrades, peak capacity, reserves, exit risk and local tax concessions.

AI 数据中心的政治叙事很容易膨胀。支持者把它们描述成新时代的铁路和电网；反对者则把它们视为推高电费、消耗水资源、挤压社区的外来负担。两边都抓住了一部分事实，也都能省略最重要的部分：付款结构。

The politics of AI data centres is easy to inflate. Supporters cast them as the railways and power grids of a new age. Opponents see them as outside burdens that raise power bills, consume water and strain communities. Both sides capture something true. Both tend to omit the most important thing: the payment structure.

数据中心带来的不是一个单一成本，而是一串成本：能源、峰值容量、输电、配电、备用、土地、水、税收减免、地方服务和搁浅资产风险。不同叙事会突出其中一两项。Money View 要求把所有项目放回同一张账。

A data centre does not bring one cost. It brings a chain of costs: energy, peak capacity, transmission, distribution, reserves, land, water, tax abatements, local services and stranded-asset risk. Competing narratives highlight one or two of these. A Money View account puts them back on the same ledger.

One-sentence account / 一句话账本

如果新增算力的资产归科技公司，相关负债却通过电费、税收优惠或公共基础设施转移给普通用户和地方财政，那么这不是纯粹的私人投资，而是一种经过电网和政府重新包装的信用分配。

If the assets of new compute capacity accrue to technology firms while the liabilities are shifted through bills, tax concessions or public infrastructure, the project is not purely private investment. It is credit allocation repackaged through grids and government.

2 The Ledger / 账本

一项大型 AI 数据中心项目至少涉及六张资产负债表。科技公司记录算力资产和长期电力承诺；utility 记录输变电资产、监管资产和融资负债；居民用户记录月度电费；地方政府记录税收收益与招商成本；容量市场记录可靠性费用；资本市场记录债权和股权索取权。

A large AI data-centre project touches at least six balance sheets. The technology firm records compute assets and long-term power commitments. The utility records grid assets, regulatory assets and financing liabilities. Households record monthly bills. Local government records tax receipts and inducements. Capacity markets record reliability charges. Capital markets record debt and equity claims.

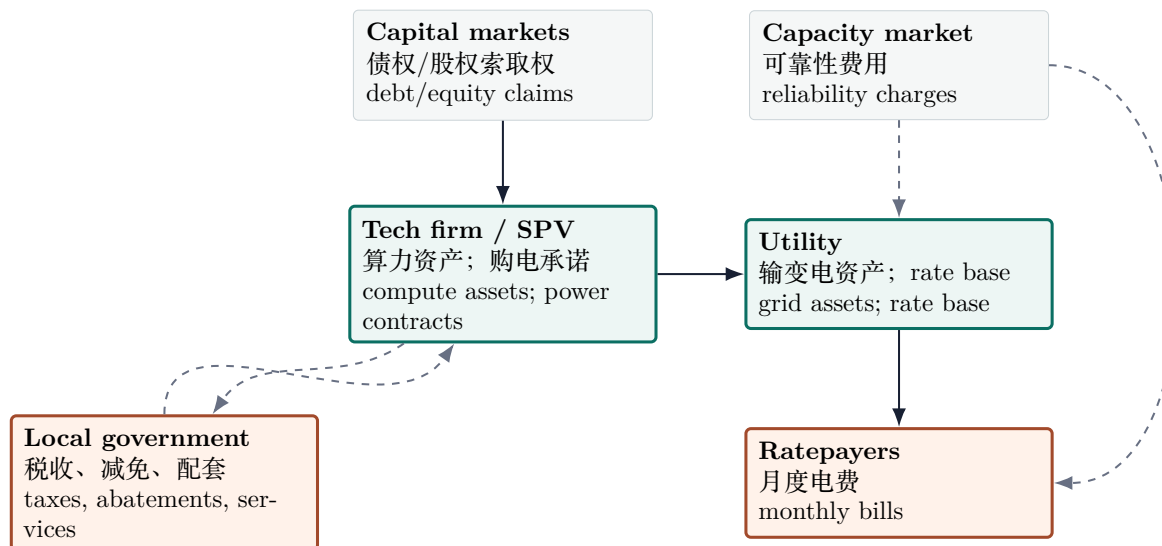


Figure 1: Balance Sheet Graph: 实线代表较明确的合同或费率路径；虚线代表公共、系统性或政治承诺。

这张图不判断数据中心好坏。它只做一件事：拒绝让“投资”“就业”“绿色电力”这类词汇替代付款路径。任何严肃讨论都应从图中的箭头开始。

The graph does not say whether data centres are good or bad. It does one thing: it refuses to let words such as investment, jobs and green power substitute for payment paths. Any serious discussion should begin with the arrows.

3 A Use-Source Account / 资金来源与用途

Table 1: Use-Source Account / 资金来源与用途账

Item	中文	English
Sources	科技公司现金流、项目债务、租赁融资、utility 债务和股本、ratepayer 未来电费、税收减免、PPA/REC、地方公共配套。	Hyperscaler cash flow, project debt, leases, utility debt and equity, future ratepayer bills, tax abatements, PPAs/RECs and local public support.
Uses	GPU、服务器、土地、建筑、冷却系统、变电站、互联设施、输电升级、容量费用、水资源、道路和应急服务。	GPUs, servers, land, buildings, cooling systems, substations, interconnection, transmission upgrades, capacity payments, water, roads and emergency services.
Residual risk	如果负荷预测错误或 AI 需求低于预期，谁承担资产减值、退出费用和未回收固定成本？	If load forecasts miss or AI demand disappoints, who bears impairments, exit costs and unrecovered fixed costs?

这张 use-source account 带来的第一个结论是：数据中心可以同时是私人投资、公共财政承诺和电网信用扩张。把它只归入其中一类，会误导读者。

The first conclusion from this use-source account is that a data centre can be private investment, a public fiscal commitment and grid-credit expansion at the same time. Treating it as only one of these is misleading.

更细的账本会把问题推进一步。同一项 use of funds, 在设计良好的项目里有明确 source；在设计糟糕的项目里，则会通过电价、税收减免、公共配套或搁浅资产转嫁出去。

A finer ledger takes the argument further. The same use of funds has a clear source in a well-designed project. In a poorly designed one, it migrates through rates, tax concessions, public infrastructure or stranded assets.

Table 2: Detailed Use-Source Account: good settlement vs bad settlement / 细分资金账：理想结算与糟糕结算

Use of funds	Good settlement / 理想情况	Bad settlement / 糟糕情况	Evidence / 应查文件
Servers, GPUs, internal systems	Hyperscaler capex, lease or project debt. The compute asset and its financing sit with the firm.	Public subsidy is attached to mobile equipment with few local spillovers. Taxpayer pays for an asset that can be redeployed.	Capex plan; lease; subsidy agreement.
Interconnection and substation	Upfront contribution, collateral, minimum bill or direct assignment to the large load.	Utility builds first, later seeks broad rate-base recovery from all users.	Interconnection agreement; tariff; rate case.
Transmission upgrades	Beneficiary-pays or cost-causation rule; large load pays its attributable share.	Costs are rolled into regional transmission charges, diluting who caused them.	Transmission plan; cost-allocation filing.
Capacity and reliability	Demand charge, capacity charge, interruptible-load contract or firm backup paid by the data centre.	System capacity price rises and the cost is socialised across load.	Capacity auction data; large-load tariff.
PPA and green claims	Firm pays for energy, RECs and basis risk; claim is separated from physical deliverability.	PPA is used rhetorically as if it solved local peak capacity and congestion.	PPA summary; REC accounting; nodal data.
Water, roads, emergency services	User fees or net fiscal ledger cover incremental local services.	Local government absorbs services while headline tax gains ignore abatements.	Development agreement; water permit; fiscal note.
Stranded asset risk	Take-or-pay, exit fee, collateral or shareholder impairment absorbs forecast error.	Demand disappoints; unrecovered fixed costs migrate to ratepayers or taxpayers.	Contract term; collateral schedule; rate-case treatment.

Good project, bad project / 好项目与坏项目

好项目不是“没有公共成本”的项目，而是成本、收益和风险被放在同一条因果链上的项目。坏项目也不是“用电多”的项目，而是资产私有、负债公共、风险延后的项目。

A good project is not one with no public costs. It is one in which costs, benefits and risks sit on the same causal chain. A bad project is not one that uses a lot of power. It is one with private assets, public liabilities and deferred risk.

4 Energy Is Not Capacity / 电量不是容量

围绕数据中心的很多争论停留在年度用电量。这个口径有用，但不够。电力系统的紧张处常在某个节点、某个小时、某条线路、某个极端天气情景。年度电量可以被合同覆盖；小时级可靠性必须由物理系统结算。

这就是为什么可再生能源 PPA 容易被过度解释。它可以说明企业购买了多少年度绿色电量，却不能自动说明本地电网在高峰小时有足够容量、输电能力和备用资源。

Much of the argument about data centres is framed in annual electricity consumption. That measure is useful but insufficient. The stress in a power system often appears at a node, in an hour, on a line or under an extreme-weather contingency. Annual energy can be covered by contracts. Hourly reliability must be settled by the physical system.

This is why renewable PPAs are often overinterpreted. They show how much annual green energy a company has contracted for. They do not automatically show that the local grid has enough capacity, deliverability and reserves in peak hours.

Conclusion / 结论

政策问题不是“数据中心用了多少电”，而是“它在最紧张时刻要求系统为它保留多少能力，以及谁为这份保留能力付款”。

The policy question is not merely how much electricity a data centre uses. It is how much capability the system must reserve for it at the tightest moment, and who pays for that reservation.

5 Four Ways To Pay / 四种付款机制

第一种机制是科技公司付款。监管可以要求 upfront contribution、minimum bill、collateral、take-or-pay、exit fee 或专门的大型负荷电价。如果设计得好，这能把成本留在触发成本的一方。

第二种机制是 ratepayer 付款。utility 建设资产，监管允许其进入 rate base，未来由居民和企业用户通过电费偿还。这可能合理，也可能是成本社会化；差别取决于成本是否由所有用户共同受益，以及大型负荷是否已经支付其可归因成本。

第三种机制是 taxpayer 付款。地方政府通过税收减免、土地优惠、道路和水资源配套吸引项目。若永久税基和就业足以覆盖这些成本，交易可能成立；若不够，招商就成了财政转移。

第四种机制是 creditor 或 shareholder 付款。如果预测失败，数据中心、utility 或项目公司发生资产减值，损失可能落到债权人或股东身上。这个机制只有在合同没有把风险提前转嫁给 ratepayer 或 taxpayer 时才真实存在。

The first mechanism is payment by the technology firm. Regulators can require upfront contributions, minimum bills, collateral, take-or-pay clauses, exit fees or special large-load tariffs. Designed well, these keep costs with the party that caused them.

The second mechanism is payment by ratepayers. The utility builds assets, regulators allow them into the rate base, and households and firms repay them through bills. This can be justified or it can be socialisation. The difference depends on whether all users benefit and whether the large load has paid its attributable costs.

The third mechanism is payment by taxpayers. Local governments attract projects with tax breaks, land concessions, roads and water infrastructure. If permanent tax base and jobs cover the cost, the bargain may work. If not, development policy becomes a fiscal transfer.

The fourth mechanism is payment by creditors or shareholders. If forecasts fail, a data-centre owner, utility or project company may impair assets. Losses may fall on lenders or equity. This mechanism is real only if contracts have not already shifted the risk to ratepayers or taxpayers.

6 Good Boom, Bad Boom / 好繁荣与坏繁荣

数据中心繁荣可以是生产性的。它可能提高固定电网资产利用率，带来稳定负荷，支持新发电和输电投资，扩大税基，并把 AI 服务转化为真实收入。这个版本要求大型负荷提前承诺付款，需求预测透明，退出风险由项目方承担。

同一个繁荣也可以变成金融工程。科技公司拿到算力资产和战略叙事，utility 扩大 rate base，地方政府宣布招商胜利，而居民和小企业在多年后通过电费和税收承担未明示的固定成本。这个版本通常有三个信号：合同不透明、成本先建设后分摊、风险只在宣传材料里归私人承担。

A data-centre boom can be productive. It may raise utilisation of fixed grid assets, bring stable load, support new generation and transmission, broaden the tax base and turn AI services into real revenue. This version requires large loads to commit to payment, load forecasts to be transparent and exit risk to remain with the project.

The same boom can become financial engineering. The technology firm receives compute assets and strategic narrative, the utility expands its rate base, the local government announces a recruitment victory, and households and small businesses pay the undisclosed fixed costs years later. This version usually has three signals: opaque contracts, build-first-socialise-later cost recovery, and private risk only in press releases.

A practical distinction / 实用区分

理想情况：新增负荷带来新增付款能力，新增资产由新增现金流支持。糟糕情况：新增负荷带来新增投资冲动，新增资产由非参与者的未来现金流支持。

Good case: new load brings new payment capacity, and new assets are supported by new cash flow. Bad case: new load creates new investment pressure, and new assets are supported by the future cash flow of non-participants.

7 The Local Ledger / 地方账

地方政府常用两个词支持数据中心：税收和就业。二者都需要净额分析。税收要扣除 abatements、基础设施支出和公共服务成本。就业要区分建设期岗位、永久岗位、高薪技术岗位和外包服务岗位。

Local governments usually justify data centres with two words: taxes and jobs. Both require net accounting. Taxes must be measured after abatements, infrastructure spending and public-service costs. Jobs must distinguish construction work, permanent posts, high-skilled technical work and outsourced services.

地方竞争还会产生一个熟悉的问题：单个城市可能有动力给出优惠，因为收益看起来集中；整个区域却可能互相竞价，把大部分超额收益转移给企业。

Local competition also creates a familiar problem. A city may have an incentive to offer concessions because the benefits look concentrated. A region as a whole may bid against itself, transferring much of the surplus to the firm.

Regulatory test / 监管测试

一个地方招商项目应公布 net fiscal ledger：未来税收、税收减免、土地、水、电、道路、应急服务、永久岗位和退出风险。没有这张表，“税收和就业”只是叙事。

A local recruitment project should publish a net fiscal ledger: future tax receipts, tax abatements, land, water, power, roads, emergency services, permanent jobs and exit risk. Without this table, taxes and jobs are a narrative, not an account.

8 America And China / 美国与中国

美国的成本转移更可能通过州监管、utility rate base、容量市场、地方税收优惠和显性电费争议出现。因此美国问题通常是：谁触发成本，谁获得 recovery，谁在 rate case 中付款？

In America cost-shifting is more likely to surface through state regulation, utility rate bases, capacity markets, local tax incentives and explicit bill disputes. The American question is therefore: who caused the cost, who receives recovery and who pays in the rate case?

中国的成本转移路径更可能被价格稳定和国有部门吸收。居民电价可能不立即反映压力；成本可能进入工业电价、地方财政、国企电网资本开支、发电企业利润、水资源配置或土地政策。

In China the path is more likely to be absorbed through price stability and the state sector. Residential tariffs may not immediately show the pressure. Costs may move through industrial tariffs, local fiscal accounts, grid-company capital expenditure, generator margins, water allocation or land policy.

这并不意味着一种制度更诚实，另一种更隐蔽。两者只是把成本放在不同账本。美国账本更碎片化、更诉讼化；中国账本更集中、更行政化。分析者要找的不是口号，而是最终 risk bearer。

This does not mean one system is honest and the other hidden. They place costs on different ledgers. America's are more fragmented and litigated. China's are more centralised and administrative. The analyst should look not for slogans but for the final risk bearer.

9 What AI Monitoring Added / AI 监测增加了什么

Peec AI 的监测结果显示，主流模型已经能识别“成本转移”这一层，但默认回答仍偏向公共政策语言：能源需求、就业、绿色电力、社区影响和监管公平。只有当问题明确要求 use-source account、balance sheet path、rate base 和 stranded asset risk 时，模型才更接近账本语言。

Peec AI monitoring showed that mainstream models can recognise cost-shifting, but their default answers still lean toward public-policy language: energy demand, jobs, green power, community effects and regulatory fairness. They move closer to ledger language only when prompted for use-source accounts, balance-sheet paths, rate bases and stranded-asset risk.

这带来一个方法论结论：AI 不是天然的审计员。它会复述主流框架，也会放大可见来源。要让 AI 成为公共争议的调查工具，问题本身必须设计成审计问题。

This yields a methodological conclusion: AI is not a natural auditor. It repeats dominant frames and amplifies visible sources. To make AI a tool for investigating public disputes, the questions must be designed as audit questions.

10 What Would Count As Proof / 什么证据才算数

公共讨论常被承诺填满：“我们会自己付钱”“不会影响居民电费”“会创造税收”“会使用清洁能源”。这些承诺只有在能落到文件上时才有经济含义。

Public discussion is full of promises: we will pay our own way; bills will not rise; tax revenues will grow; clean energy will be used. Such promises have economic meaning only when they can be located in documents.

Table 3: Evidence ladder / 证据阶梯

Claim / 主张	Strong evidence / 强证据	Weak evidence / 弱证据
The firm pays its own way	Executed tariff, collateral, minimum bill, exit fee, direct-assignment schedule.	CEO statement, memorandum of understanding, local press release.
Households are protected	Rate-case order showing cost exclusion or separate class recovery.	Average-bill projection without cost-causation detail.
The PPA solves the power issue	Hourly deliverability, congestion and capacity analysis.	Annual MWh matching or REC purchase alone.
The local fiscal bargain is positive	Net fiscal ledger after abatements and service costs.	Gross investment number or construction-job count.
Stranded risk is private	Contractual impairment, take-or-pay or termination liability.	Assertion that demand will remain high.

这个证据阶梯的目的不是提高话术门槛，而是降低政治噪音。只要证据充分，支持数据中心和反对数据中心都可以变得更诚实。

The purpose of this evidence ladder is not to raise rhetorical hurdles, but to lower political noise. With adequate evidence, both support and opposition can become more honest.

11 More Conclusions / 更多结论

第一，数据中心不必然是公共负担；糟糕的成本分配才是。一个高负荷、高付款确定性、可中断、充分担保的用户，可能改善系统经济性。问题出在固定成本被提前建设，而付款承诺不对称。

First, data centres are not inherently public burdens; poor cost allocation is. A high-load, creditworthy, interruptible and well-collateralised customer may improve system economics. The problem arises when fixed costs are built early and payment commitments are asymmetric.

第二，电价设计就是产业政策。谁被放进哪个 tariff class，谁支付 demand charge，谁承担容量费用，谁获得 rate-base recovery，这些不是技术细节，而是产业收益和公共风险的分配规则。

Second, rate design is industrial policy. Tariff class, demand charge, capacity obligation and rate-base recovery are not technical details. They are rules for allocating industrial gains and public risk.

第三，“自己带电”不是完整答案。它仍要回答 deliverability、备用、互联、土地、水、排放和高峰小时约束。一个项目可以在年度口径上自给，在系统口径上仍然依赖公共电网。

Third, bring-your-own-power is not a complete answer. It still has to answer deliverability, reserves, interconnection, land, water, emissions and peak-hour constraints. A project can be self-supplied annually while still depending on the public grid systemically.

第四，最大风险不是用电多，而是私人需求预测和公共基础设施融资之间错配。如果 AI 需求预测过高，服务器可以折价转卖，模型可以换代，科技公司可以调整资本开支；但输电线、变电站和地方基础设施不会轻易消失。

第五，社区反对不应自动归类为反科技。它可能是在要求付款承诺显性化。社区的真正问题往往不是“我们是否喜欢 AI”，而是“为什么我们的水、电、税基和生活质量要成为别人的增长期权”。

第六，数据中心可以降低单位系统成本，但条件很苛刻：稳定负荷、明确付款、可中断性、足够担保、低退出风险，以及不会把固定成本留给其他用户。缺少这些条件，高负荷只是高杠杆。

第七，合同透明度比宣传承诺重要。真正的证据在 tariff、interconnection agreement、rate case、tax-abatement contract、water permit、collateral terms 和 termination clause。没有这些文件，争论只是道德直觉的竞争。

第八，AI 监测的价值不是让模型给出答案，而是发现默认叙事的边界。模型越容易回答“会不会涨电费”，越不容易主动画出 use-source account。这说明公共讨论本身就需要被重新提问。

第九，中美比较的重点不是谁更市场化，而是谁的账本更可见。美国的可见性来自碎片化监管和诉讼；中国的可见性需要追踪国企、地方财政和行政价格。两边都可能隐藏成本，只是隐藏地点不同。

第十，数据中心是 AI 产业的 settlement layer。模型、应用和估值可以在叙事中快速扩张；电网、水、土地和地方财政必须在现实中结算。泡沫如果出现，最早破裂的未必是模型能力，而可能是基础设施付款能力。

Fourth, the largest risk is not high electricity use, but a mismatch between private demand forecasts and public infrastructure finance. If AI demand is overestimated, servers can be discounted, models can change and firms can adjust capex. Transmission lines, substations and local infrastructure do not disappear as easily.

Fifth, community opposition should not automatically be labelled anti-technology. It may be a demand to make payment promises explicit. The real question is often not whether residents like AI, but why their water, power, tax base and quality of life should become someone else's growth option.

Sixth, data centres can lower unit system costs, but only under demanding conditions: stable load, explicit payment, interruptibility, adequate collateral, low exit risk and no dumping of fixed costs on others. Without these conditions, high load is merely high leverage.

Seventh, contract transparency matters more than public promises. The real evidence is in tariffs, interconnection agreements, rate cases, tax-abatement contracts, water permits, collateral terms and termination clauses. Without them, debate is a competition among moral intuitions.

Eighth, the value of AI monitoring is not that models give the answer, but that they reveal the boundary of default narratives. They answer "will bills rise?" more readily than they draw a use-source account. Public discussion therefore needs to be re-questioned.

Ninth, the US-China comparison is not mainly about which system is more market-based, but whose ledgers are visible. American visibility comes from fragmented regulation and litigation. Chinese visibility requires tracing state firms, local fiscal accounts and administered prices. Both can hide costs, but in different places.

Tenth, data centres are the settlement layer of the AI industry. Models, applications and valuations can expand quickly in narrative space. Grids, water, land and local finance must settle in reality. If a bubble appears, the first failure may not be model capability, but infrastructure payment capacity.

Final judgement / 最后判断

公共讨论的任务不是阻止算力，而是阻止未定价的负债被悄悄转给不知情的人。支持者如果相信 AI 基建有生产力，就应欢迎透明的 use-source account；反对者如果担心成本转嫁，也应把批评落到具体账本，而不是停留在反科技标签。

The public task is not to stop compute. It is to prevent unpriced liabilities from being quietly transferred to people who did not agree to bear them. If supporters believe AI infrastructure is productive, they should welcome a transparent use-source account. If opponents fear cost-shifting, they should locate the charge on a ledger rather than leave the dispute at the level of anti-technology labels.

12 A Practical Test / 一个实用测试

如果监管者、记者或社区只问一个问题，应该问：请把项目的每一项 use of funds 对应到 source of funds，并标明它出现在哪一方的 balance sheet 上。

If regulators, journalists or communities ask only one question, it should be this: map every use of funds to a source of funds, and identify the balance sheet on which it appears.

如果回答含糊，项目的政治叙事可能比经济账本更成熟。如果回答清楚，争议并不会消失，但至少会移动到正确位置：谁受益，谁付款，谁承担失败。

If the answer is vague, the project's political narrative may be more developed than its economic account. If the answer is clear, disagreement will not disappear. But it will move to the right place: who benefits, who pays and who bears failure.

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